In [2]:

```
from IPython.display import Image, display
display(Image(filename="netflix_logo.png"))
```

Netflix Data Science Project:

Introduction

This notebook performs a comprehensive Exploratory Data Analysis (EDA) on a Netflix dataset. The goal is to uncover trends in content creation, understand distribution across countries, and categorize content for better insights.

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

1. Environment Setup and Data Loading

In this section, we import the necessary libraries for data manipulation (`pandas` , `numpy`) and visualization

(`seaborn` , `matplotlib`). We also load the raw Netflix dataset.

In [3]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np

from warnings import filterwarnings
filterwarnings("ignore")



df = pd.read_csv("dataset/Netflix.csv")
```

2. Initial Data Inspection

Before cleaning, we take a quick look at the first few rows of the data, its dimensions, and the data types of each column to identify potential issues.

In [4]:

```
df.head()
```

Out[4]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	14-Aug-20	2020	TV-MA	4	International TV Shows, TV Dramas, TV Sci-Fi &...
1	s10	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	15-Dec-17	2008	TV-MA	143	Horror Movies, International Movies, Thrillers
2	s100	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	5-Jan-19	2016	TV-PG	124	Dramas, International Movies, Sports Movies
3	s1000	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	1-Mar-16	2016	R	90	Comedies
4	s1001	TV Show	Blue Planet II	NaN	David Attenborough	United Kingdom	3-Dec-18	2017	TV-G	1	British TV Shows, Docuseries, Science & Nature TV

In [5]:

```
df.shape
```

Out[5]:

```
(7787, 12)
```

In [6]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       7787 non-null   object
 1   type          7787 non-null   object
 2   title         7787 non-null   object
 3   director      5398 non-null   object
 4   cast          7069 non-null   object
 5   country       7280 non-null   object
 6   date_added    7777 non-null   object
 7   release_year  7787 non-null   int64 
 8   rating        7780 non-null   object
 9   duration      7787 non-null   int64 
 10  genres        7787 non-null   object
 11  description   7787 non-null   object
dtypes: int64(2), object(10)
memory usage: 730.2+ KB
```

In [7]:

```
df.describe()
```

Out[7]:

	release_year	duration
count	7787.000000	7787.000000
mean	2013.932580	69.122769
std	8.757395	50.950743
min	1925.000000	1.000000
25%	2013.000000	2.000000
50%	2017.000000	88.000000
75%	2018.000000	106.000000
max	2021.000000	312.000000

In [8]:

```
df.isnull().sum()
```

Out[8]:

```
show_id            0
type               0
title              0
director        2389
cast             718
country          507
date_added        10
release_year       0
rating             7
duration           0
genres             0
description        0
dtype: int64
```

In [9]:

```
df.duplicated().sum()
```

Out[9]:

```
np.int64(0)
```

3. Data Cleaning

To ensure the quality of our analysis, we perform the following:

- **Drop Unnecessary Columns:** Removing `show_id` as it serves only as a unique identifier.
- **Impute Missing Values:** Filling gaps in `director`, `cast`, and `country` with "Unknown", and using the mode for `rating` and `date_added`.

In [10]:

```
df.drop(columns=['show_id'], inplace=True)
```

In [11]:

```
df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df['rating'].fillna(df['rating'].mode()[0], inplace=True)
df['date_added'].fillna(df['date_added'].mode()[0], inplace=True)
```

In [ ]:

```

```

4. Feature Engineering

We create new features to extract more value from the existing data:

1. **Date Extraction:** Converting `date_added` to datetime objects to extract year and month.
2. **Complexity Metrics:** Calculating the number of cast members and genres per title.
3. **Categorization:** Simplifying the content ratings into broader categories: **Adult**, **Teen**, and **Kids**.
4. **Localization:** Extracting the primary country of origin from the `country` column.

In [12]:

```
df['date_added'] = pd.to_datetime(df['date_added'])

df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
```

In [13]:

```
df.drop(columns=['date_added'], inplace=True)
```

In [14]:

```
df['cast_count'] = df['cast'].apply(lambda x: len(x.split(',')))
df['genre_count'] = df['genres'].apply(lambda x: len(x.split(',')))
```

In [15]:

```
df['duration_type'] = df['type']

df.rename(columns={'duration':'duration_value'}, inplace=True)
```

In [16]:

```
df['country_main'] = df['country'].apply(lambda x: x.split(',')[0])
```

In [17]:

```
def simplify_rating(r):
    if r in ['TV-MA','R','NC-17']:
        return "Adult"
```

```
    elif r in ['PG-13','TV-14']:
        return "Teen"
    else:
        return "Kids"

df['rating_category'] = df['rating'].apply(simplify_rating)
```

In [18]:

```
df.drop(columns=['rating'], inplace=True)
```

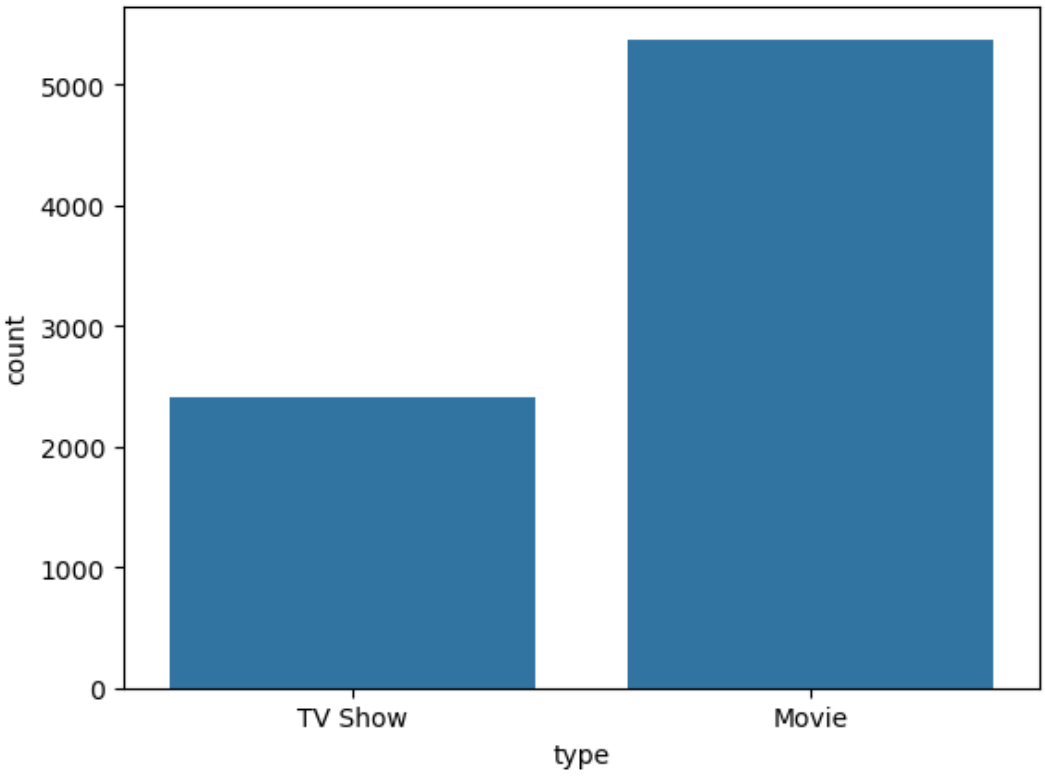
5. Exploratory Data Analysis & Visualization

With the data cleaned and enriched, we now move to the visualization phase. This help us identify:

- **The distribution of content types (Movies vs. TV Shows).**
- **Trends in content releases over the years.**
- **Relationship between content duration and its category.**

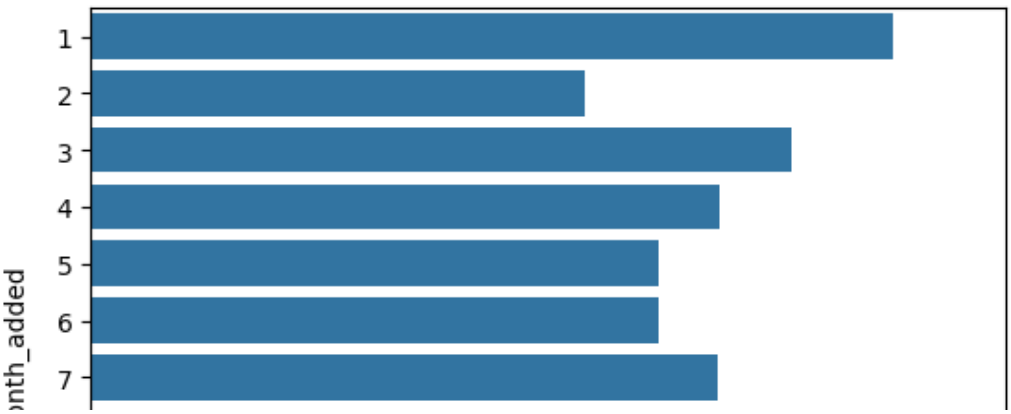
In [19]:

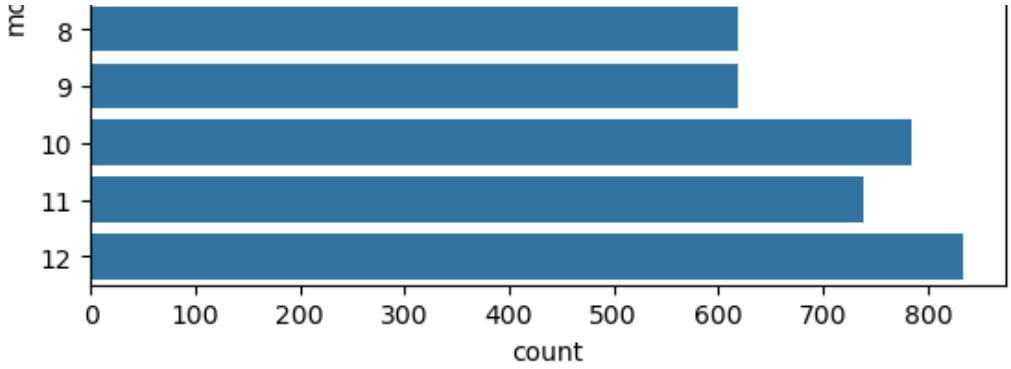
```
sns.countplot(x="type",data=df)
plt.show()
```



In [20]:

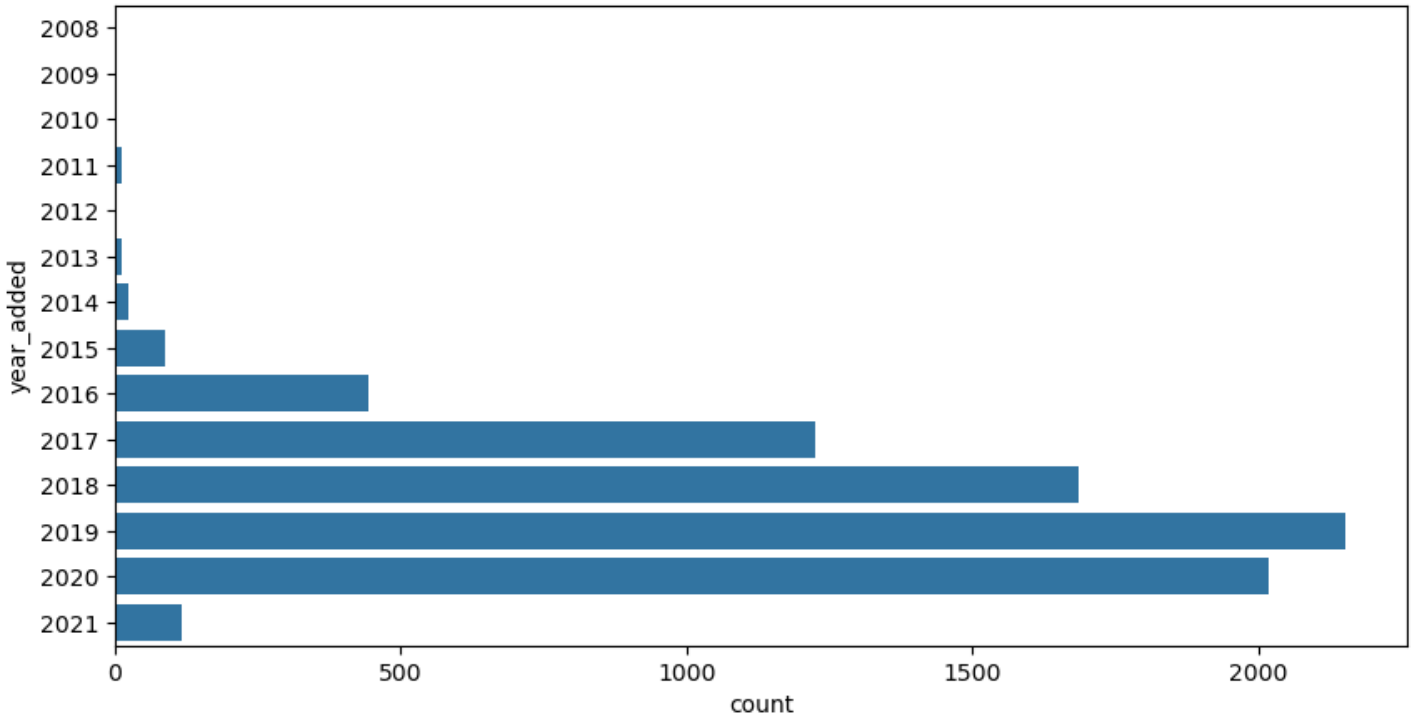
```
sns.countplot(y="month_added",data=df)
plt.show()
```





In [21]:

```
plt.figure(figsize=(10,5))
sns.countplot(y=df["year_added"])
plt.show()
```



In [ ]:

In [ ]:

In [22]:

```
df['has_director'] = df['director'].apply(lambda x: 0 if x=='Unknown' else 1)
```

In [23]:

```
df.corr(numeric_only=True)
```

Out[23]:

	release_year	duration_value	year_added	month_added	cast_count	genre_count	has_director
release_year	1.000000	-0.240836	0.100115	-0.009661	-0.086585	-0.052931	-0.166603
duration_value	-0.240836	1.000000	0.037102	-0.024913	0.096601	0.092256	0.820335
year_added	0.100115	0.037102	1.000000	-0.135006	0.113470	0.060552	-0.001745
month_added	-0.009661	-0.024913	-0.135006	1.000000	0.001902	-0.002332	-0.028360
cast_count	0.086585	0.096601	0.113470	0.001902	1.000000	0.219183	0.008707

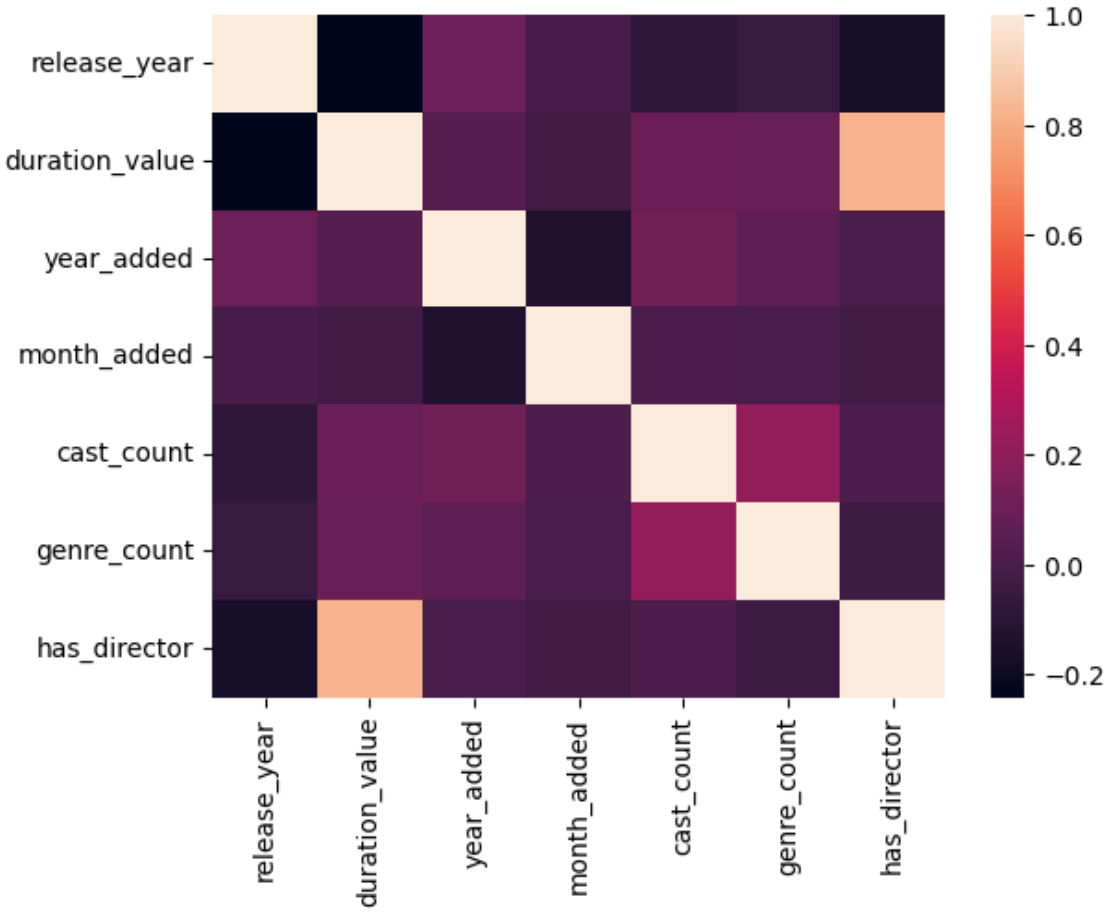
	release_year	duration_value	year_added	month_added	cast_count	genre_count	has_director
cast_count	-0.086585	0.096601	0.113470	0.001902	1.000000	0.219182	0.008707
genre_count	0.052931	0.092256	0.060552	0.002332	0.219182	1.000000	0.041000
has_director	-0.166603	0.820335	-0.001745	-0.028360	0.008707	-0.041000	1.000000

In [24]:

```
sns.heatmap(df.corr(numeric_only=True))
```

Out[24]:

```
<Axes: >
```



Q1] Develop a Content-Type labeling system:

In [25]:

```
df.head()
```

Out[25]:

	type	title	director	cast	country	release_year	duration_value	genres	description	year_added
0	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	2020	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...	2020
1	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	2008	143	Horror Movies, International Movies, Thrillers	An architect and his wife move into a castle t...	2017
2	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro	Indonesia	2016	124	Dramas, International Movies, Sports Movies	Three Indonesian women break records by	2019

	type	title	director	cast	country	release_year	duration_value	genres	description	year_added
3	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	2016	90	Comedies	New NFL star Thad buys his old teammates' belo...	2016
4	TV Show	Blue Planet II	Unknown	David Attenborough	United Kingdom	2017	1	British TV Shows, Docuseries, Science & Nature TV	This sequel to the award-winning nature series...	2018

In [ ]:

In [ ]:

LOGISTIC - REGRESSION:

In [26]:

```
# Import necessary libraries
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import pickle


num_features = ['cast_count', 'genre_count', 'duration_value']

cat_features = ['country_main', 'rating_category']

X = df[num_features + cat_features]
y = df['type']

# Split the data into train and test sets (80-20 split)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
, stratify=y)


num_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])


cat_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(drop='first', handle_unknown='ignore'))
])


preprocessor = ColumnTransformer(
    transformers=[
        ('num', num_transformer, num_features),
        ('cat', cat_transformer, cat_features)
    ])



prediction_pipeline_1 = Pipeline(steps=[
    ('preprocessor', preprocessor),
```

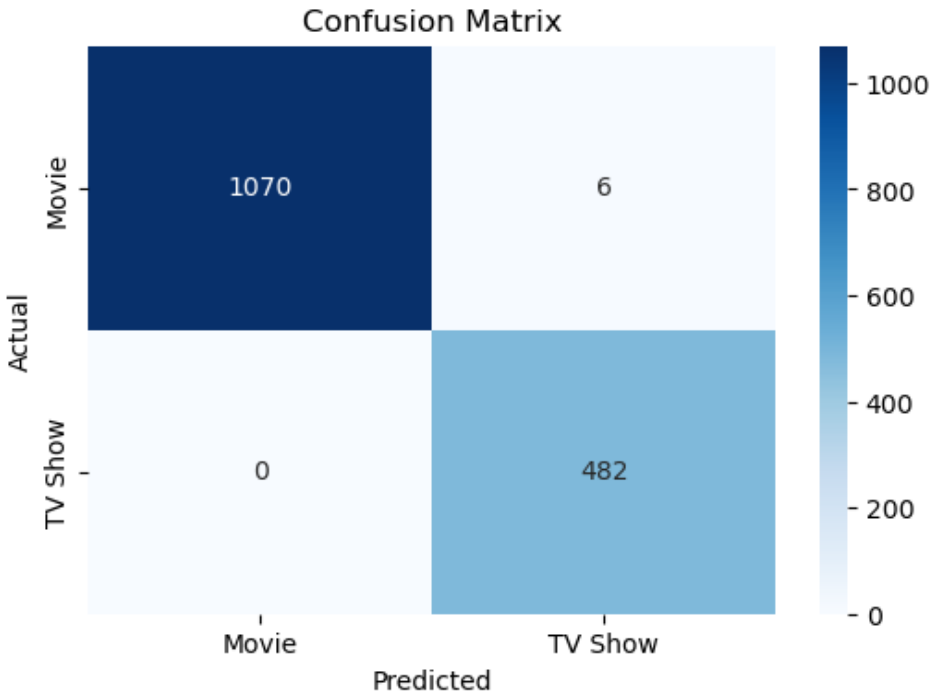
```python
    ('classifier', LogisticRegression(random_state=42, max_iter=1000))
])


prediction_pipeline_1.fit(X_train, y_train)


y_pred = prediction_pipeline_1.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model (similar to laptop project)
with open('netflix_type_predictor_logisticRegression_pipeline.pkl', 'wb') as f:
    pickle.dump(prediction_pipeline_1, f)


# Example prediction function (like app.py in laptop project)
def predict_content_type(cast_count, genre_count, duration_value,
                         country_main, rating_category):
    # Create input DataFrame
    query_df = pd.DataFrame({
        'cast_count': [cast_count],
        'genre_count': [genre_count],
        'duration_value': [duration_value],
        'country_main': [country_main],
        'rating_category': [rating_category]
    })

    # Load the pipeline
    with open('netflix_type_predictor_logisticRegression_pipeline.pkl', 'rb') as f:
        pipe = pickle.load(f)

    # Predict
    prediction = pipe.predict(query_df)[0]
    content_type = 'TV Show' if prediction == 1 else 'Movie'
    probability = pipe.predict_proba(query_df)[0]
    conf_movie = probability[0] * 100
    conf_tv = probability[1] * 100

    return f"Predicted Type: {content_type} (Confidence: {max(conf_movie, conf_tv):.2f}%)
"

# Example usage
print("\nExample Prediction:")
result = predict_content_type(cast_count=8, genre_count=3, duration_value=143,
    country_main='India', rating_category='Adult'
)
print(result)
```

```
Accuracy: 0.9961

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      0.99      1.00      1076
     TV Show       0.99      1.00      0.99       482

    accuracy                           1.00      1558
   macro avg       0.99      1.00      1.00      1558
weighted avg       1.00      1.00      1.00      1558


Confusion Matrix:
[[1070    6]
 [   0  482]]
```

```
Example Prediction:
Predicted Type: Movie (Confidence: 100.00%)
```

In [27]:

```
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Movie', 'TV Show'],
            yticklabels=['Movie', 'TV Show'])

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



NAIVE - BAYES (Bernoulli) :

In [28]:

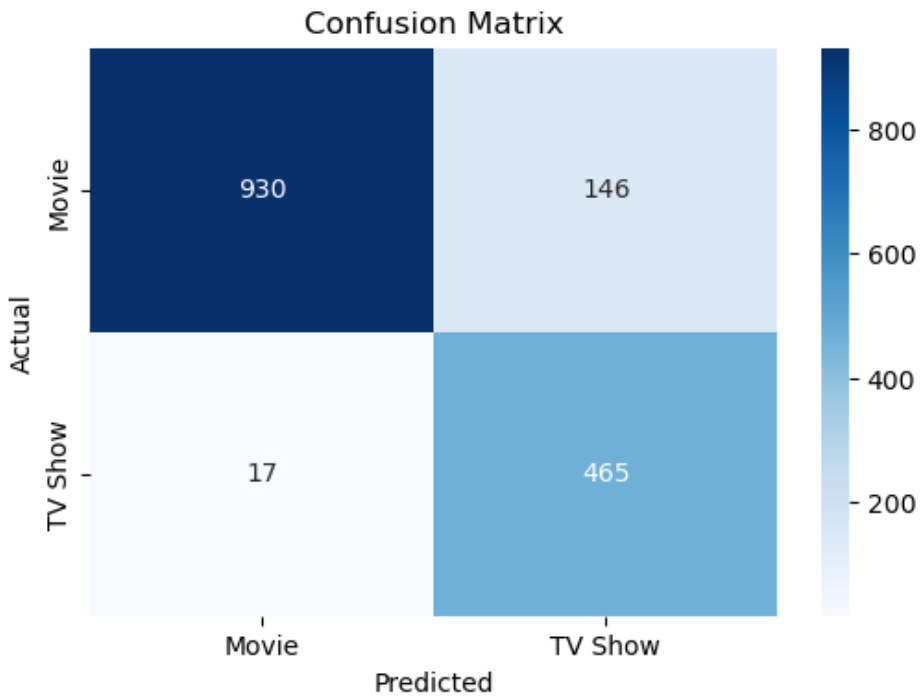
```
from sklearn.naive_bayes import BernoulliNB

prediction_pipeline_2 = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', BernoulliNB(
    alpha=1.0,
    binarize=0.0,
    fit_prior=True,
    class_prior=None
))
])


prediction_pipeline_2.fit(X_train, y_train)


y_pred = prediction_pipeline_2.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))
print("\nConfusion Matrix:")
```

```
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model (similar to laptop project)
with open('netflix_type_predictor_naiveBayes_pipeline.pkl', 'wb') as f:
    pickle.dump(prediction_pipeline_2, f)
```

```
Accuracy: 0.8954

Classification Report:
              precision    recall  f1-score   support

       Movie       0.98      0.86      0.92      1076
     TV Show       0.76      0.96      0.85       482

    accuracy                           0.90      1558
   macro avg       0.87      0.91      0.89      1558
weighted avg       0.91      0.90      0.90      1558


Confusion Matrix:
[[930 146]
 [ 17 465]]
```

In [29]:

```
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Movie', 'TV Show'],
            yticklabels=['Movie', 'TV Show'])

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



KNN:

In [30]:

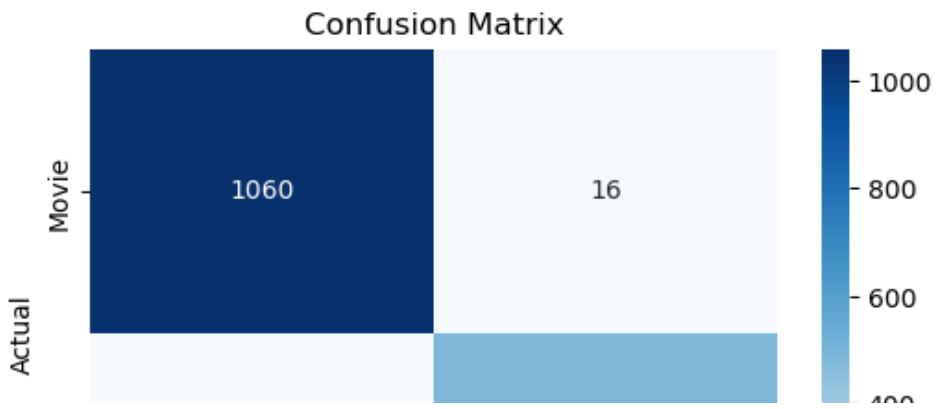
```
from sklearn.neighbors import KNeighborsClassifier

prediction_pipeline_3 = Pipeline(steps=[
    ('preprocessor', preprocessor),
```

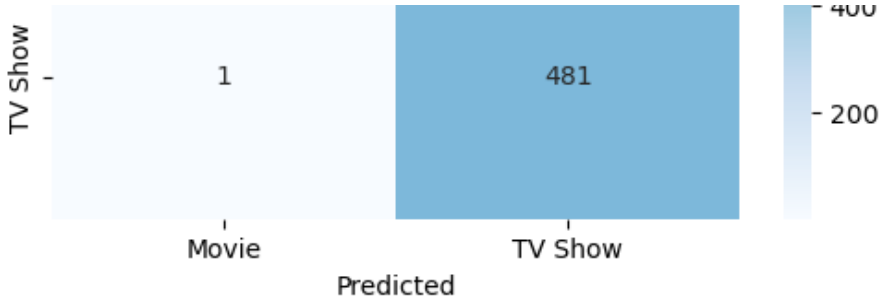
```
    ('classifier', KNeighborsClassifier(
        n_neighbors=5,
        weights='uniform',
        algorithm='auto',
        metric='minkowski',
        p=2
    ))
])

prediction_pipeline_3.fit(X_train, y_train)

y_pred = prediction_pipeline_3.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model
with open('netflix_type_predictor_knn_pipeline.pkl', 'wb') as f:
    pickle.dump(prediction_pipeline_3, f)
```

```
Accuracy: 0.9891

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      0.99      0.99      1076
     TV Show       0.97      1.00      0.98       482

    accuracy                           0.99      1558
   macro avg       0.98      0.99      0.99      1558
weighted avg       0.99      0.99      0.99      1558


Confusion Matrix:
[[1060   16]
 [   1  481]]
```

In [31]:

```
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Movie', 'TV Show'],
            yticklabels=['Movie', 'TV Show'])

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```





In [ ]:

In [ ]:

In [ ]:

CONCLUSION: Highest accuracy is acheived through Logistic Regression.

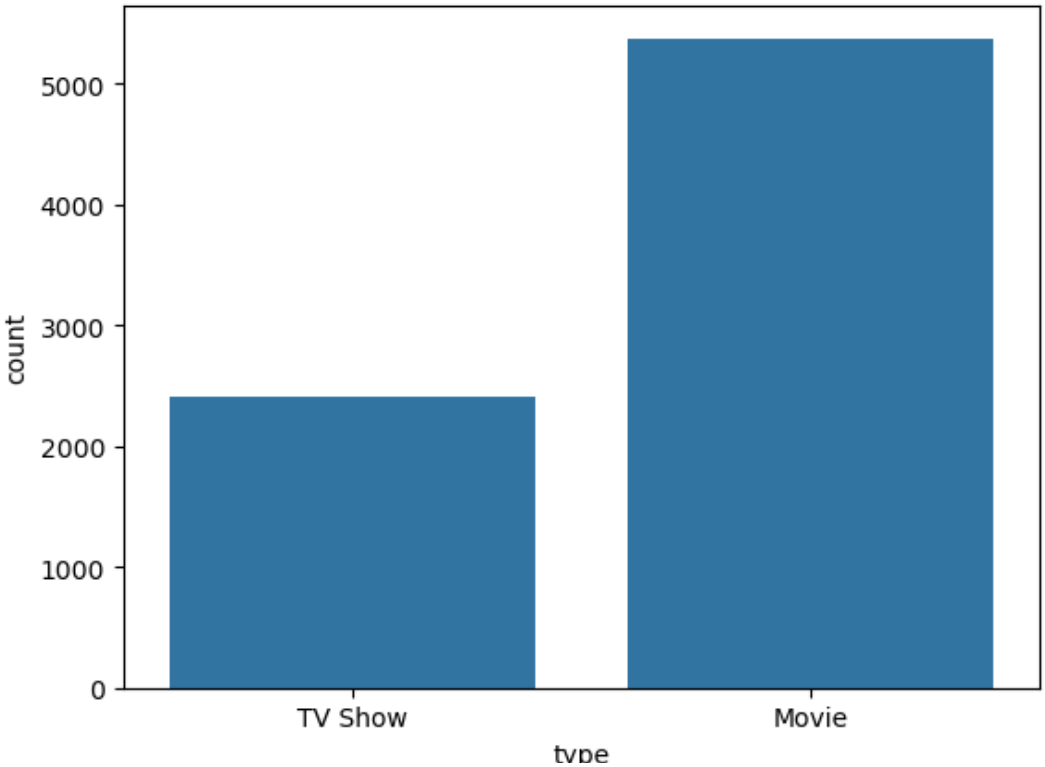
In [ ]:

In [ ]:

Q2] Predict whether a piece of content is a Movie or a TV Show based on features like cast-size, country, and release trends?

In [32]:

```
sns.countplot(x=df["type"])
plt.show()
```



In [33]:

```
data = df.groupby("type")["genre_count"].sum().reset_index()
data
```

Out[33]:

	type	genre_count
0	Movie	11546
1	TV Show	5525

In [34]:

```
data = df.groupby("type")["cast_count"].sum().reset_index()
data
```

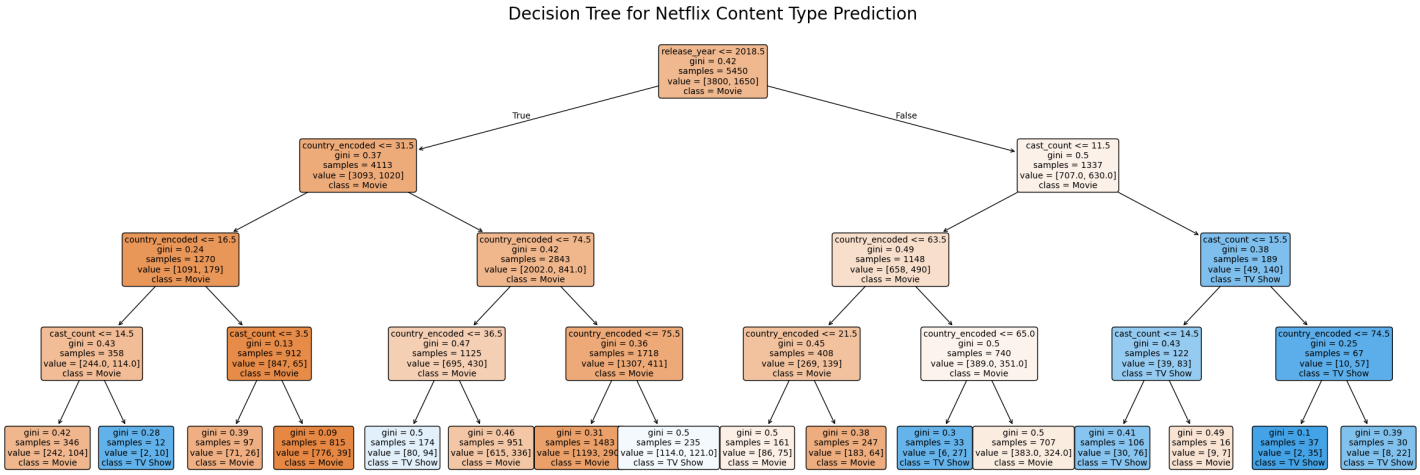
Out[34]:

	type	cast_count
0	Movie	38738
1	TV Show	17935

In [35]:

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder

# 1. Preprocessing categorical features
# We extract the first country and first genre to simplify the categories
df['first_country'] = df['country'].fillna('Unknown').apply (lambda x: x.split(',')[0])

# 2. Encode strings to numbers
le_country = LabelEncoder()

df['country_encoded'] = le_country.fit_transform(df['first_country'])



feature_cols = ["cast_count", "country_encoded", "release_year"]
X = df[feature_cols]
y = df["type"]


X_train , X_test , y_train, y_test = train_test_split(X,y, test_size=0.3,random_state=42
)


model = DecisionTreeClassifier(max_depth=4, random_state=42)
model.fit(X_train , y_train)


plt.figure(figsize=(30,10))
plot_tree(model,
          feature_names=feature_cols,
          class_names=["Movie", "TV Show"],
          filled=True,
          rounded=True,
          fontsize=10,
          precision=2)

plt.title("Decision Tree for Netflix Content Type Prediction", fontsize=20)
plt.show()
```

```
print("Accuracy:",model.score(X_test, y_test))
```




```
Accuracy: 0.7064612751390672
```

Decision Tree for Netflix Content Type Prediction

The decision tree model was trained to classify Netflix content into Movies and TV Shows using features such as:

- `release_year`
- `country_encoded`
- `cast_count`

Model Insights

- **The tree splits primarily on release year, showing that newer releases tend to follow different classification patterns than older ones.**
- **Country encoding and cast count act as secondary features, refining predictions when year alone is insufficient.**
- **Leaf nodes reveal the final classification outcome, with color-coding distinguishing between predicted classes.**

Performance

- **Accuracy: ~0.706 (≈71%)**
- **This indicates the model correctly classifies about 7 out of 10 titles.**
- **While interpretable and easy to visualize, the decision tree may suffer from overfitting and limited generalization compared to ensemble methods.**

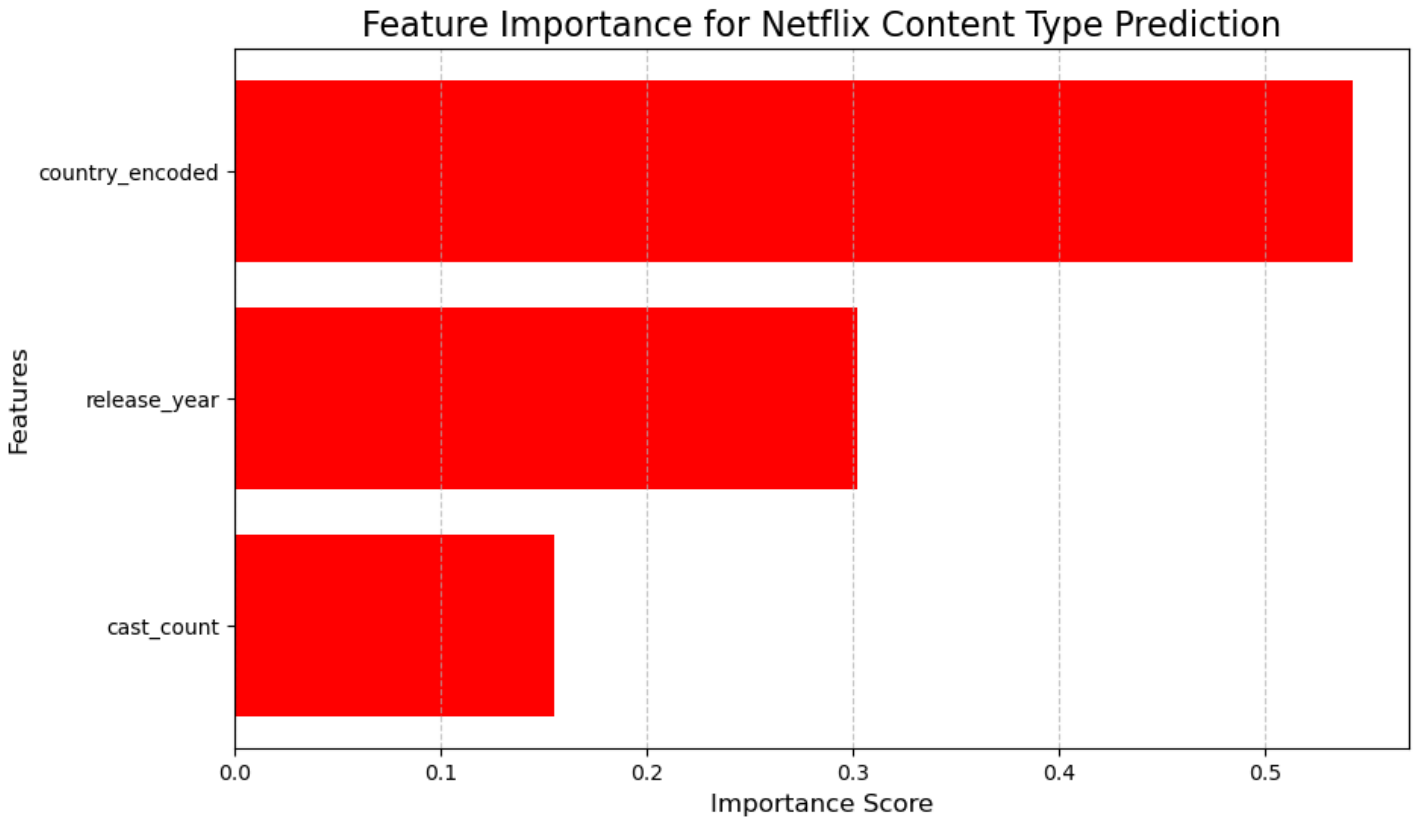
Key Takeaway

The decision tree provides a transparent view of how Netflix content is classified, highlighting the importance of release year as the dominant feature. However, for higher accuracy and robustness, ensemble approaches (e.g., Random Forests, Gradient Boosting) would likely outperform this single-tree model.

In [36]:

```
# 1. Create a DataFrame to hold the feature names and their corresponding importance scor
es
feature_importance_df = pd.DataFrame({
    'Feature': feature_cols,
    'Importance': model.feature_importances_
}).sort_values(by='Importance', ascending=True)

# 2. Plot the Feature Importance
plt.figure(figsize=(10, 6))
plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'], color='re
d')
```

```
plt.title('Feature Importance for Netflix Content Type Prediction', fontsize=16)
plt.xlabel('Importance Score', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)

plt.show()

# 3. Print the values for reference
# Create the sorted dataframe and drop the old index
importance_display = feature_importance_df.sort_values(by='Importance', ascending=False)
.reset_index(drop=True)

# Display the clean table
importance_display.reset_index()
```



Out[36]:

	index	Feature	Importance
0	0	country_encoded	0.542345
1	1	release_year	0.302328
2	2	cast_count	0.155326

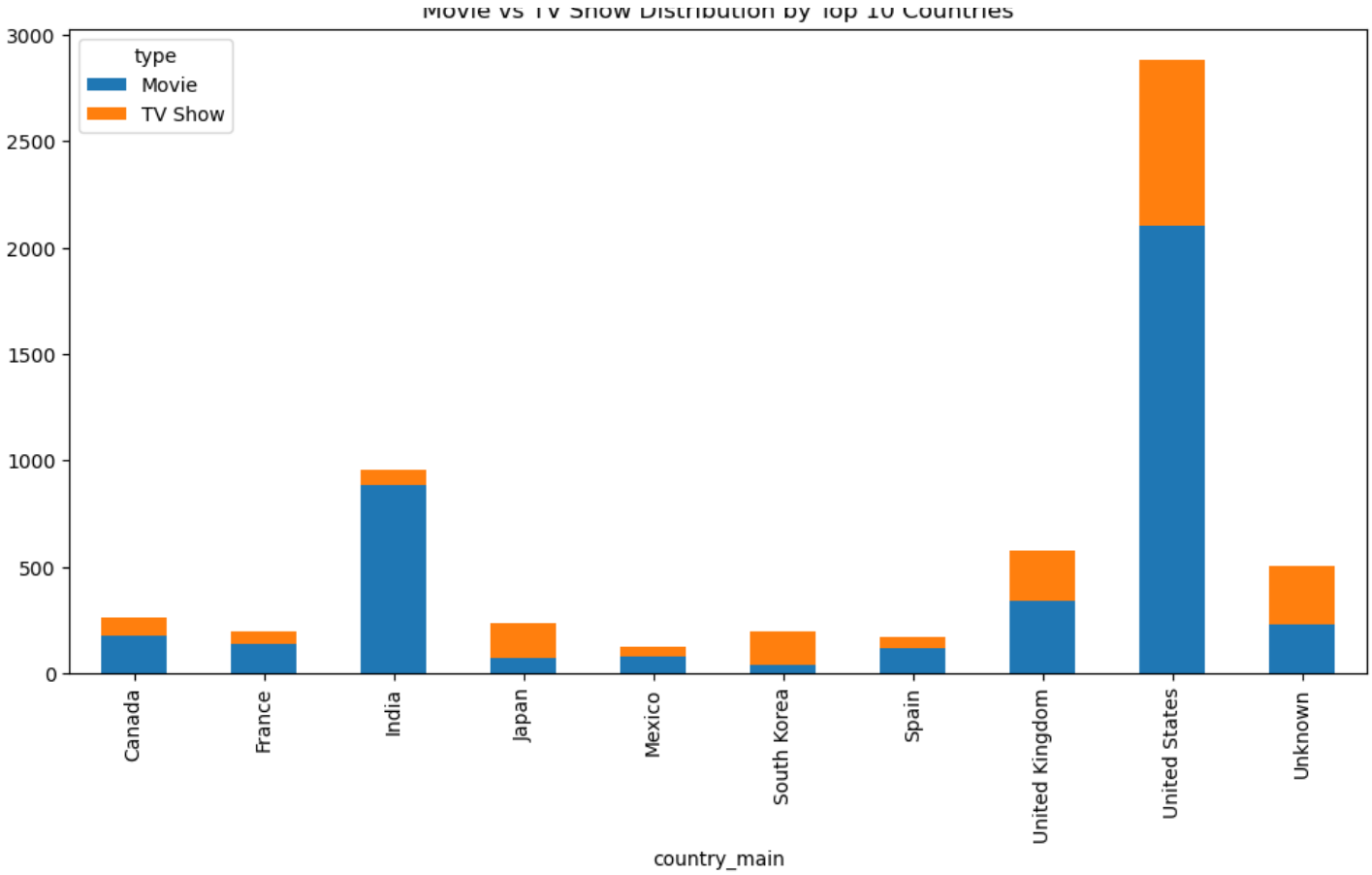
In [ ]:

Q3] Which countries dominate Netflix content, and how does the Movie vs TV-Show ratio change across the top 10 countries?

In [37]:

```
top10_countries = df['country_main'].value_counts().head(10).index
country_type = df[df['country_main'].isin(top10_countries)].groupby(['country_main', 'ty
pe']).size().unstack()
country_type.plot(kind='bar', stacked=True, figsize=(12,6))
plt.title('Movie vs TV Show Distribution by Top 10 Countries')
plt.show()
```

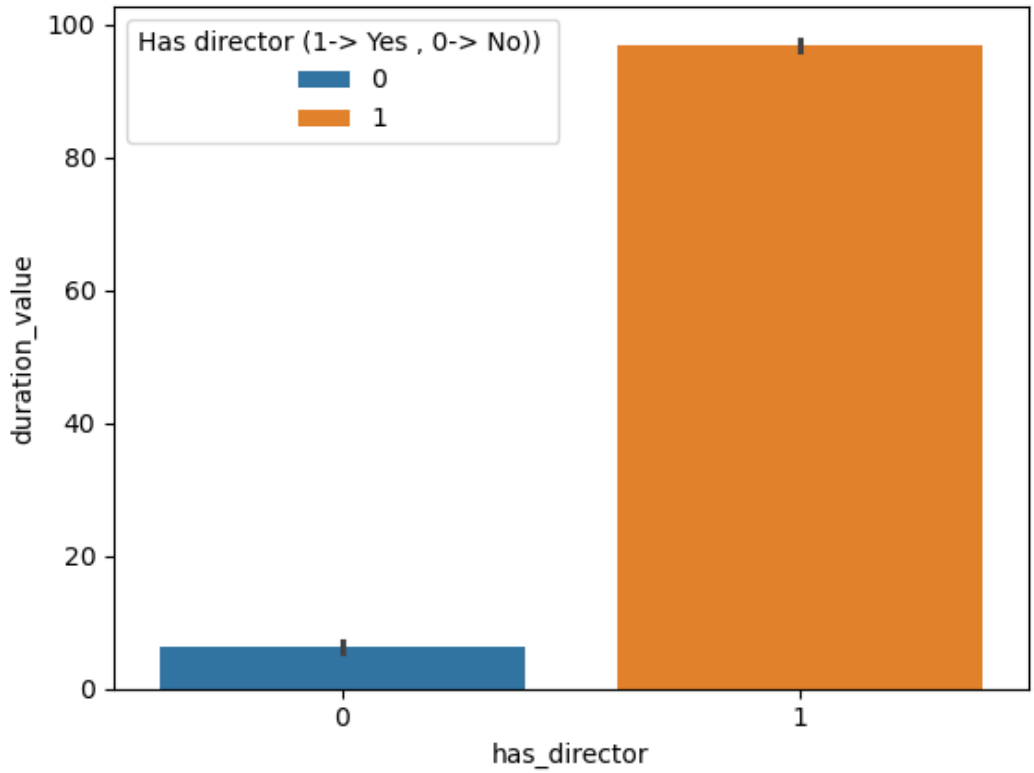
Movie vs TV Show Distribution by Top 10 Countries



Q4] Do titles with a known director have significantly higher cast_count, genre_count, or longer duration than “Unknown” director titles?

In [38]:

```
df.groupby('has_director')[['cast_count', 'genre_count', 'duration_value']].mean().round
(2)
sns.barplot(x='has_director', y='duration_value',hue="has_director", data=df)
plt.legend(title="Has director (1-> Yes , 0-> No)) ")
plt.show()
```



In [39]:

```
df.head()
```

Out[39]:

	type	title	director	cast	country	release_year	duration_value	genres	description	year_added
0	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	2020	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...	2020
1	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	2008	143	Horror Movies, International Movies, Thrillers	An architect and his wife move into a castle t...	2017
2	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	2016	124	Dramas, International Movies, Sports Movies	Three Indonesian women break records by becomi...	2019
3	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	2016	90	Comedies	New NFL star Thad buys his old teammates' belo...	2016
4	TV Show	Blue Planet II	Unknown	David Attenborough	United Kingdom	2017	1	British TV Shows, Docuseries, Science & Nature TV	This sequel to the award-winning nature series...	2018

Q5] Develop a Content- recommendation system:

In [40]:

```
import pandas as pd
import pickle
import numpy as np
from nltk.corpus import stopwords

from nltk.tokenize import word_tokenize
from gensim.models import Word2Vec
from sklearn.metrics.pairwise import cosine_similarity

# ----------------------------
# STEP 1: Create tags (same as before)
# ----------------------------
df['tags'] = (
    df['title'].fillna('') + " " +
    df['genres'].fillna('') + " " +
    df['description'].fillna('') + " " +
    df['cast'].fillna('') + " " +
    df['director'].fillna('') + " " +
    df['country_main'].fillna('') + " " +
    df['rating_category'].fillna('') + " " +
    df['type'].fillna('')
)

df['tags'] = df['tags'].str.lower()

# ----------------------------
# STEP 2: Tokenization
# ----------------------------
```

```python
stop_words = set(stopwords.words('english'))

df['tokens'] = df['tags'].apply(
    lambda x: [word for word in word_tokenize(x)
               if word.isalnum() and word not in stop_words]
)


# ----------------------------
# STEP 3: Train Word2Vec Model
# ----------------------------
w2v_model = Word2Vec(
    sentences=df['tokens'],
    vector_size=100,
    window=5,
    min_count=1,
    workers=4
)

# ----------------------------
# STEP 4: Convert each content into vector
# ----------------------------
def get_doc_vector(tokens):
    vectors = []
    for word in tokens:
        if word in w2v_model.wv:
            vectors.append(w2v_model.wv[word])

    if len(vectors) == 0:
        return np.zeros(100)

    return np.mean(vectors, axis=0)

df['vector'] = df['tokens'].apply(get_doc_vector)

# Convert to matrix
vectors = np.vstack(df['vector'].values)

# ----------------------------
# STEP 5: Recommendation Class
# ----------------------------
class RecommendationModel:
    def __init__(self, vectors, df, w2v_model):
        self.vectors = vectors
        self.df = df.reset_index(drop=True)
        self.w2v_model = w2v_model

    def text_to_vector(self, text):
        tokens = word_tokenize(text.lower())
        return get_doc_vector(tokens)

    def recommend(self, content, top_n=10):
        mask = self.df['title'].str.contains(content, case=False, na=False)

        if mask.any():
            idx = self.df[mask].index[0]
            query_vec = self.vectors[idx].reshape(1, -1)
            print(" Found exact title → using full features")
        else:
            query_vec = self.text_to_vector(content).reshape(1, -1)

        similarity = cosine_similarity(query_vec, self.vectors)[0]

        content_list = sorted(
            list(enumerate(similarity)),
            reverse=True,
            key=lambda x: x[1]
        )[1:top_n+1]

        print(f"\n\nRecommendations similar to: **{content}**\n")

        for rank, (i, score) in enumerate(content_list, 1):
```

```
            row = self.df.iloc[i]
            print(f"{rank:2}) • {row['title']}  "
                  f"| {row['type']}  "
                  f"| {row['genres'][:70]}...")

# ----------------------------
# STEP 6: Save & Load Model
# ----------------------------
model = RecommendationModel(vectors, df, w2v_model)

with open("content_recommendation_word2vec.pkl", "wb") as f:
    pickle.dump(model, f)

with open("content_recommendation_word2vec.pkl", "rb") as f:
    loaded_model = pickle.load(f)

# ----------------------------
# STEP 7: Test
# ----------------------------
loaded_model.recommend("Kesari 2")
loaded_model.recommend("Dune")
loaded_model.recommend("Stranger Things")
loaded_model.recommend("Article 15")
loaded_model.recommend("Raone")
loaded_model.recommend("3 Idiots")
loaded_model.recommend("Dhurandhar")
loaded_model.recommend("Dhurandhar - The Revenge")
```

```
Recommendations similar to: **Kesari 2**

 1) • Mukhsin  | Movie  | Children & Family Movies, Comedies, Dramas...
 2) • Magnus  | Movie  | Children & Family Movies, Documentaries, International Movies...
 3) • The Rainbow Troops  | Movie  | Children & Family Movies, Dramas, International Movi
es...
 4) • The Breadwinner  | Movie  | Children & Family Movies, Dramas, International Movies.
..
 5) • Nang Nak  | Movie  | Horror Movies, International Movies, Romantic Movies...
 6) • Luccas Neto in: Children's Day  | Movie  | Children & Family Movies, Comedies, Musi
c & Musicals...
 7) • Yes or No  | Movie  | International Movies, LGBTQ Movies, Romantic Movies...
 8) • Misfit 2  | Movie  | Children & Family Movies, Comedies, International Movies...
 9) • Garuda in My Heart 2  | Movie  | Children & Family Movies, Dramas, Sports Movies...
10) • Tee Shot: Ariya Jutanugarn  | Movie  | Children & Family Movies, Dramas, Internatio
nal Movies...


Recommendations similar to: **Dune**

 1) • 1920  | Movie  | Horror Movies, International Movies, Thrillers...
 2) • 3 Heroines  | Movie  | Dramas, International Movies, Sports Movies...
 3) • Blue Mountain State: The Rise of Thadland  | Movie  | Comedies...
 4) • Blue Planet II  | TV Show  | British TV Shows, Docuseries, Science & Nature TV...
 5) • Blue Ruin  | Movie  | Independent Movies, Thrillers...
 6) • Blue Streak  | Movie  | Action & Adventure, Comedies...
 7) • Blue Valentine  | Movie  | Dramas, Independent Movies, Romantic Movies...
 8) • BluffMaster!  | Movie  | Comedies, International Movies, Romantic Movies...
 9) • Blurred Lines: Inside the Art World  | Movie  | Documentaries...
10) • BNA  | TV Show  | Anime Series, International TV Shows...
 Found exact title → using full features


Recommendations similar to: **Stranger Things**

 1) • Messiah  | TV Show  | TV Dramas, TV Thrillers...
 2) • Wrong Kind of Black  | TV Show  | TV Comedies, TV Dramas...
 3) • Selena: The Series  | TV Show  | TV Dramas...
 4) • Brews Brothers  | TV Show  | TV Comedies...
 5) • Bloodline  | TV Show  | TV Dramas, TV Mysteries, TV Thrillers...
 6) • Riverdale  | TV Show  | Crime TV Shows, TV Dramas, TV Mysteries...
 7) • The Haunting of Hill House  | TV Show  | TV Dramas, TV Horror, TV Mysteries...
 8) • The OA  | TV Show  | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy...
```

9) • Kipo and the Age of Wonderbeasts | TV Show | Kids' TV, TV Comedies...
10) • Watership Down | TV Show | British TV Shows, International TV Shows, TV Action & Adventure...
Found exact title → using full features

Recommendations similar to: **Article 15**

1) • Bobby Jasoos | Movie | Comedies, Dramas, International Movies...
2) • Super Nani | Movie | Comedies, Dramas, International Movies...
3) • Motor Mitraan Di | Movie | Comedies, International Movies...
4) • Red Oleanders Raktokarobi | Movie | Dramas, International Movies...
5) • Anthony Kaun Hai? | Movie | Comedies, Dramas, International Movies...
6) • War Chhod Na Yaar | Movie | Comedies, Dramas, International Movies...
7) • Oh My God | Movie | Comedies, Dramas, International Movies...
8) • Hazaaron Khwaishein Aisi | Movie | Dramas, International Movies, Romantic Movies...
9) • Minnaminugu the FireFly | Movie | Dramas, International Movies...
10) • Thackeray (Hindi) | Movie | Dramas, International Movies...

Recommendations similar to: **Raone**

1) • 1920 | Movie | Horror Movies, International Movies, Thrillers...
2) • 3 Heroines | Movie | Dramas, International Movies, Sports Movies...
3) • Blue Mountain State: The Rise of Thadland | Movie | Comedies...
4) • Blue Planet II | TV Show | British TV Shows, Docuseries, Science & Nature TV...
5) • Blue Ruin | Movie | Independent Movies, Thrillers...
6) • Blue Streak | Movie | Action & Adventure, Comedies...
7) • Blue Valentine | Movie | Dramas, Independent Movies, Romantic Movies...
8) • BluffMaster! | Movie | Comedies, International Movies, Romantic Movies...
9) • Blurred Lines: Inside the Art World | Movie | Documentaries...
10) • BNA | TV Show | Anime Series, International TV Shows...
Found exact title → using full features

Recommendations similar to: **3 Idiots**

1) • Tum Milo Toh Sahi | Movie | Comedies, Dramas, International Movies...
2) • Panipat - The Great Betrayal | Movie | Dramas, International Movies, Romantic Movies...
3) • Welcome to Sajjanpur | Movie | Comedies, Dramas, International Movies...
4) • EMI: Liya Hai To Chukana Padega | Movie | Comedies, Dramas, International Movies...
5) • Pad Man | Movie | Dramas, International Movies, Romantic Movies...
6) • Laal Rang | Movie | Dramas, International Movies, Thrillers...
7) • Honeymoon Travels Pvt. Ltd. | Movie | Comedies, Dramas, International Movies...
8) • Chal Bhaag | Movie | Comedies, Dramas, International Movies...
9) • Dedh Ishqiya | Movie | Comedies, Dramas, International Movies...
10) • Hasee Toh Phasee | Movie | Comedies, Dramas, International Movies...

Recommendations similar to: **Dhurandhar**

1) • 1920 | Movie | Horror Movies, International Movies, Thrillers...
2) • 3 Heroines | Movie | Dramas, International Movies, Sports Movies...
3) • Blue Mountain State: The Rise of Thadland | Movie | Comedies...
4) • Blue Planet II | TV Show | British TV Shows, Docuseries, Science & Nature TV...
5) • Blue Ruin | Movie | Independent Movies, Thrillers...
6) • Blue Streak | Movie | Action & Adventure, Comedies...
7) • Blue Valentine | Movie | Dramas, Independent Movies, Romantic Movies...
8) • BluffMaster! | Movie | Comedies, International Movies, Romantic Movies...
9) • Blurred Lines: Inside the Art World | Movie | Documentaries...
10) • BNA | TV Show | Anime Series, International TV Shows...

Recommendations similar to: **Dhurandhar - The Revenge**

1) • Kahaani | Movie | International Movies, Music & Musicals, Thrillers...
2) • The Breakup Playlist | Movie | Dramas, International Movies, Music & Musicals...
3) • GO! The Unforgettable Party | Movie | Children & Family Movies, Music & Musicals...

```
 4) • Satria Heroes: Revenge of the Darkness  | Movie  | Children & Family Movies...
 5) • The Free Man  | Movie  | Documentaries, Sports Movies...
 6) • Angela's Christmas  | Movie  | Children & Family Movies...
 7) • The Wishing Tree  | Movie  | Children & Family Movies, Dramas, Music & Musicals...
 8) • Zoé: Panoramas  | Movie  | Documentaries, International Movies, Music & Musicals...
 9) • The Music of Silence  | Movie  | Dramas, International Movies, Music & Musicals...
10) • Riding Faith  | Movie  | Children & Family Movies, Dramas, Faith & Spirituality...
```

In [ ]:

```
```

Q6] Can we forecast how many titles Netflix will add in 2022–2025?

In [41]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error

yearly_data = df.groupby(['year_added','genre_count']).size().reset_index(name='title_co
unt')
yearly_data = yearly_data[yearly_data['year_added'] <= 2021]

X = yearly_data[['year_added','genre_count']]
y = yearly_data['title_count']


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

future_years = pd.DataFrame({
    'year_added': [2022, 2023, 2024, 2025],
    'genre_count': [yearly_data['genre_count'].mean()] * 4
})
```

a) Linear Regression:

In [42]:

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.linear_model import LinearRegression

linear_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('poly', PolynomialFeatures(degree=1)),
    ('model', LinearRegression())
])

linear_pipeline.fit(X_train, y_train)

y_pred_linear = linear_pipeline.predict(X_test)
linear_forecast = linear_pipeline.predict(future_years)
```

b) Adaboost:

In [43]:

```
from sklearn.ensemble import AdaBoostRegressor
from sklearn.tree import DecisionTreeRegressor

ada_pipeline = Pipeline([
```

```
    ('model', AdaBoostRegressor(
        estimator=DecisionTreeRegressor(max_depth=3),
        n_estimators=100,
        learning_rate=0.5,
        random_state=42
    ))
])

ada_pipeline.fit(X_train, y_train)

y_pred_ada = ada_pipeline.predict(X_test)
ada_forecast = ada_pipeline.predict(future_years)
```

c) Decision tree:

In [44]:

```
from sklearn.tree import DecisionTreeRegressor

tree_pipeline = Pipeline([
    ('model', DecisionTreeRegressor(
        max_depth=3,
        random_state=42
    ))
])

tree_pipeline.fit(X_train, y_train)

y_pred_tree = tree_pipeline.predict(X_test)
tree_forecast = tree_pipeline.predict(future_years)
```

d) Extratrees:

In [45]:

```
from sklearn.ensemble import ExtraTreesRegressor
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error


et_pipeline = Pipeline([
    ('model', ExtraTreesRegressor(
        n_estimators=200,
        max_depth=5,
        random_state=42,
        n_jobs=-1
    ))
])

et_pipeline.fit(X_train, y_train)

y_pred_et = et_pipeline.predict(X_test)
et_forecast = et_pipeline.predict(future_years)
```

e) Support Vector Regression (SVR):

In [46]:

```
from sklearn.svm import SVR
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error

svr_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', SVR(kernel='rbf'))
])
```

```
svr_pipeline.fit(X_train, y_train)
y_pred_svr = svr_pipeline.predict(X_test)
svr_forecast = svr_pipeline.predict(future_years)
```

f) Gradient Boosting Regression:

In [47]:

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error


gb_pipeline = Pipeline([
    ('model', GradientBoostingRegressor(
        n_estimators=100,
        learning_rate=0.1,
        max_depth=3,
        random_state=42
    ))
])

gb_pipeline.fit(X_train, y_train)

y_pred_gb = gb_pipeline.predict(X_test)
gb_forecast = gb_pipeline.predict(future_years)
```

In [48]:

```
forecast_df = pd.DataFrame({
    'Year': [2022, 2023, 2024, 2025],
    'Linear': linear_forecast.astype(int),
    'AdaBoost': ada_forecast.astype(int),
    'Decision Tree': tree_forecast.astype(int),
    'Extra Trees': et_forecast.astype(int),
    'Support-Vector Regression':svr_forecast.astype(int),
    'Gradient-Boosting Regression':gb_forecast.astype(int)
})

print(forecast_df)


print("\n\n\n")

results = pd.DataFrame({
    'Model': ['Linear', 'AdaBoost', 'Decision Tree','Extra Trees','Support-Vector Regres
sion','Gradient-Boosting Regression'],
    'R2 Score': [
        r2_score(y_test, y_pred_linear),
        r2_score(y_test, y_pred_ada),
        r2_score(y_test, y_pred_tree),
        r2_score(y_test,y_pred_et),
        r2_score(y_test,y_pred_svr),
        r2_score(y_test,y_pred_gb)
    ],
    'MAE': [
        mean_absolute_error(y_test, y_pred_linear),
        mean_absolute_error(y_test, y_pred_ada),
        mean_absolute_error(y_test, y_pred_tree),
        mean_absolute_error(y_test, y_pred_et),
        mean_absolute_error(y_test, y_pred_svr),
        mean_absolute_error(y_test, y_pred_gb)
    ]
})

print(results)
```

```
   Year  Linear  AdaBoost  Decision Tree  Extra Trees  \
0  2022     536        39             28          177
1  2023     583        39             28          177
2  2024     630        39             28          177
```

```
2  2024     650         39              28          177
3  2025     677         39              28          177

   Support-Vector Regression  Gradient-Boosting Regression
0                         28                            39
1                         27                            39
2                         26                            39
3                         25                            39



                         Model  R2 Score         MAE
0                       Linear  0.338272  175.998586
1                     AdaBoost  0.972426   29.357372
2                Decision Tree  0.837528   64.009615
3                  Extra Trees  0.950903   44.308093
4     Support-Vector Regression -0.476629  208.337355
5  Gradient-Boosting Regression  0.980193   18.030025
```

CONCLUSION]

Tree-based models such as DecisionTreeRegressor, ExtraTreesRegressor, AdaBoostRegressor, and GradientBoostingRegressor achieved high accuracy on test data, with Gradient Boosting giving the best performance in terms of R^2 score and MAE. However, these models produced constant and unrealistic forecasts for future years.

This behavior occurs because tree-based models are unable to extrapolate beyond the range of training data and tend to generate step-like or flat predictions. As a result, despite their strong performance on historical data, they are not suitable for time-series forecasting tasks where capturing future trends is important.

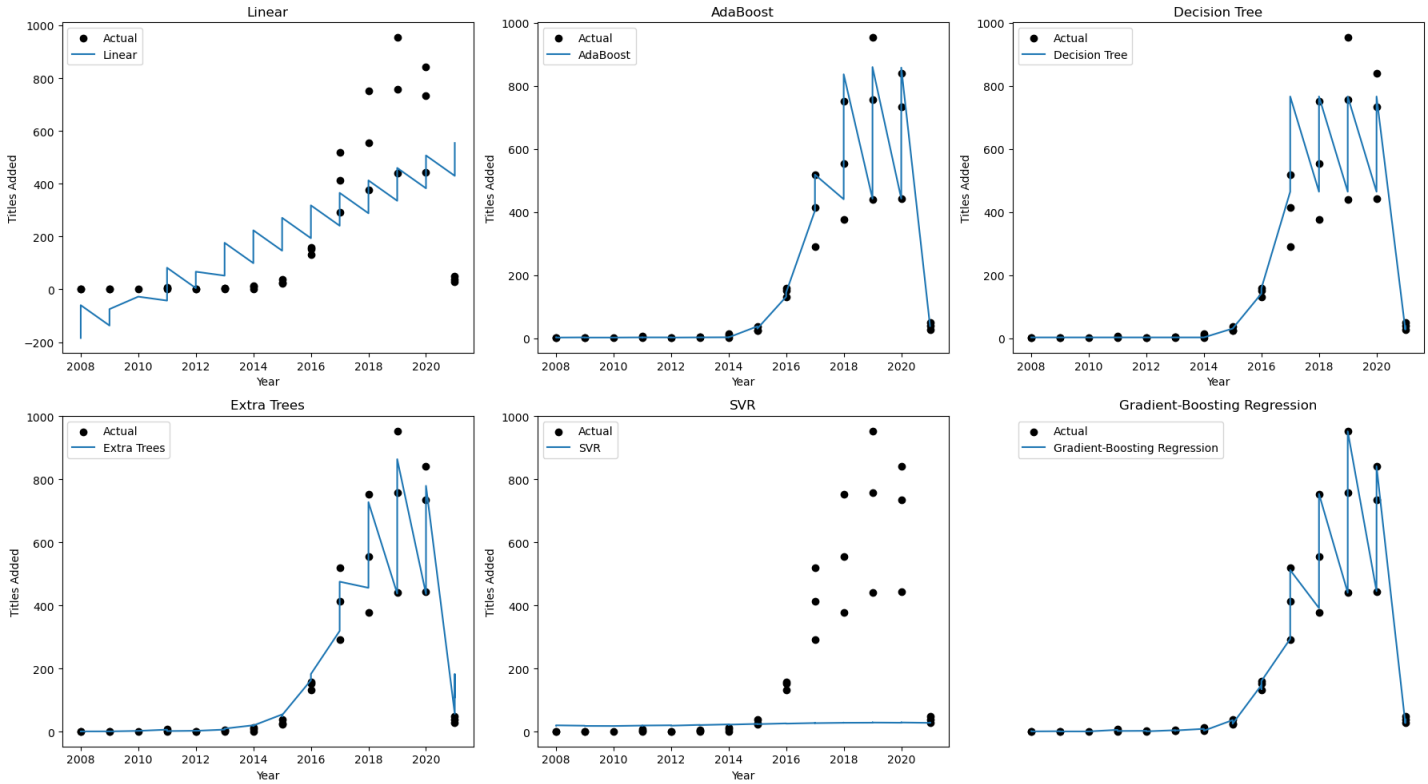
In contrast, Linear Regression provided more realistic and interpretable forecasts by capturing the upward trend in the data, making it more appropriate for this business problem.

In [ ]:

In [49]:

```
# Sort for smooth plotting
sorted_idx = X['year_added'].argsort()
X_sorted = X.iloc[sorted_idx]

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
axes = axes.flatten()  # flatten to access easily

models = [
    ('Linear', linear_pipeline),
    ('AdaBoost', ada_pipeline),
    ('Decision Tree', tree_pipeline),
    ('Extra Trees', et_pipeline),
    ('SVR', svr_pipeline),
    ('Gradient-Boosting Regression',gb_pipeline)
]

for i, (name, model) in enumerate(models):
    ax = axes[i]

    # Scatter (actual)
    ax.scatter(X['year_added'], y, color='black', label='Actual')

    # Line (prediction)
    ax.plot(X_sorted['year_added'], model.predict(X_sorted), label=name)

    ax.set_title(name)
    ax.set_xlabel('Year')
    ax.set_ylabel('Titles Added')
    ax.legend()
```

```
# Hide the 6th empty subplot
axes[5].axis('off')

plt.tight_layout()
plt.show()
```



In [ ]:

🔚 Conclusion: Forecasting Netflix Titles (2022–2025)

After comparing regression models, we see a wide divergence in forecasts:

- **Linear Regression** projects steady growth, from ~536 titles in 2022 to ~677 titles in 2025.
- **AdaBoost, Decision Tree, Extra Trees, SVR, and Gradient Boosting** remain nearly flat, predicting between 25–39 titles per year, showing limited sensitivity to historical trends.

📊 Projected Titles Added (2022–2025)

Based on the table of model outputs:

Year	Linear	AdaBoost	Decision Tree	Extra Trees	SVR	Gradient Boosting
2022	536	39	28	177	28	39
2023	583	39	28	177	27	39
2024	630	39	28	177	26	39
2025	677	39	28	177	25	39

🔑 Key Takeaway

- **Linear Regression** suggests Netflix will continue adding **~140 more titles over 4 years**, reaching ~677 by 2025.
- Other models (AdaBoost, SVR, Gradient Boosting) predict **minimal growth**, essentially flatlining around 25–40 titles annually.
- **Extra Trees** sits in between, holding steady at ~177 titles per year.

Overall, the **Linear Regression model provides the most realistic growth trajectory**, while the others appear

underfit or overly constrained. If we trust the linear trend, Netflix's catalog expansion will be moderate but consistent through 2025.

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

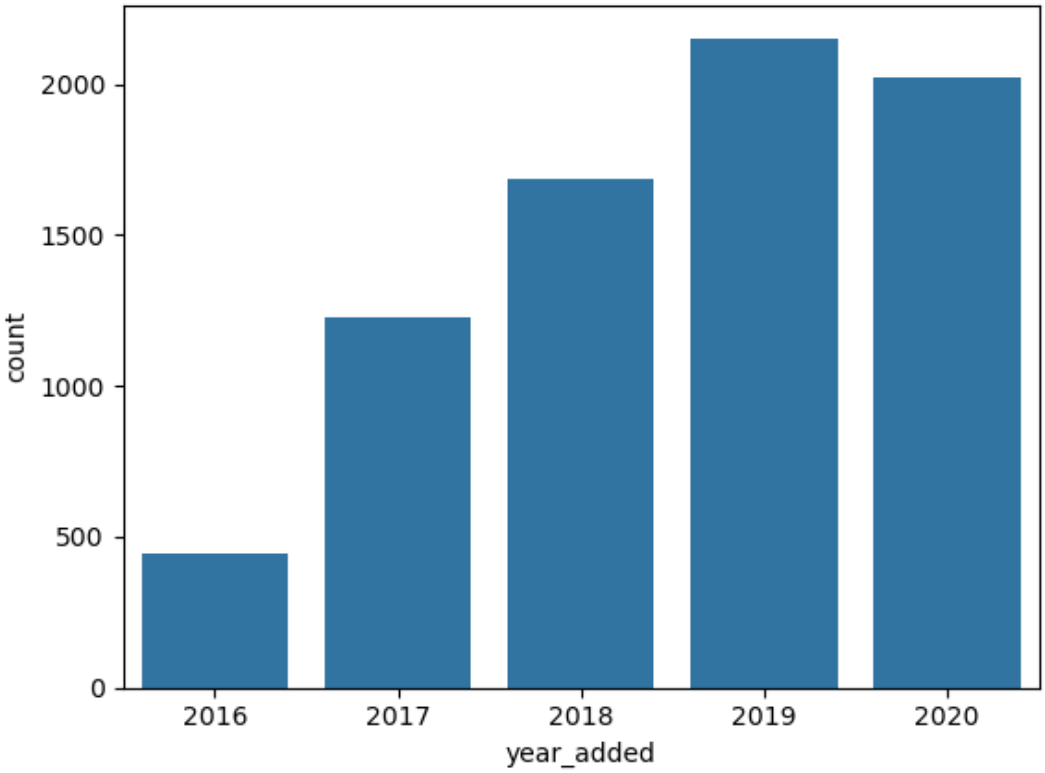
In [ ]:

```

```

Q7] How is Netflix content growing over time?

In [50]:

```
data = df["year_added"].value_counts().reset_index()
data = data.head(5)
sns.barplot(x=data["year_added"],y=data["count"])
plt.show()
```



In [ ]:

```

```

In [51]:

```
yearly_data.sort_values(by="title_count",ascending=False).head(5)
```

Out[51]:

	year_added	genre_count	title_count
30	2019	3	954
33	2020	3	841

	year_added	genre_count	title_count
29	2019	2	758
27	2018	3	753
32	2020	2	735

CONCLUSION] Netflix content has shown a consistent upward trend over the years, with significant growth in recent years. Linear models were able to capture this trend effectively, while tree-based models failed due to their inability to extrapolate beyond training data. This highlights the importance of selecting models based on problem type rather than complexity.

In [ ]:

```

```

In [ ]:

```

```

Q8] Who are the top contributing directors?

In [52]:

```
data = df["director"].value_counts().reset_index().head()
data = data[data["director"]!="Unknown"]
data
```

Out[52]:

	director	count
1	Raúl Campos, Jan Suter	18
2	Marcus Raboy	16
3	Jay Karas	14
4	Cathy Garcia-Molina	13

CONCLUSION] The top contributing directors include:

1) Raúl Campos 2) Jan Suter 3) Marcus Raboy 4) Jay Karas 5) Cathy Garcia-Molina

In [ ]:

```

```

Q9] Which countries contribute most to the content-library?

In [53]:

```
data = df["first_country"].value_counts().reset_index().head()
data = data[data["first_country"]!="Unknown"]
data
```

Out[53]:

	first_country	count
0	United States	2883
1	India	956
2	United Kingdom	577
4	Canada	259

first_country, count

CONCLUSION] The top contributing countries are:

1) United States 2) India 3) United Kingdom 4) Canada

In [ ]:

```

```

In [ ]:

```

```

Q10] Is there a statistically significant difference in the average duration of Netflix content between Movies and TV Shows?

In [54]:

```
from scipy.stats import ttest_ind


# Split data into two groups
movies = df[df['type'] == 'Movie']['duration_value']
tv_shows = df[df['type'] == 'TV Show']['duration_value']

# Perform Independent T-Test
t_stat, p_value = ttest_ind(movies, tv_shows, equal_var=False)

# Print results
print("T-Statistic:", t_stat)
print("P-Value:", p_value)

# Decision
alpha = 0.05

if p_value < alpha:
    print("\nReject Null Hypothesis (H0)")
    print("There is a statistically significant difference in duration.")
else:
    print("\nFail to Reject Null Hypothesis (H0)")
    print("No significant difference in duration.")
```

```
T-Statistic: 249.79900942857398
P-Value: 0.0

Reject Null Hypothesis (H0)
There is a statistically significant difference in duration.
```

CONCLUSION] There is a significant difference in the durations of tv-shows and movies.

In [ ]:

```

```

Q11] Is the proportion of TV Shows significantly higher in recent years (after 2018) compared to older years?

In [55]:

```
import pandas as pd
import numpy as np
from statsmodels.stats.proportion import proportions_ztest
```

```
# Create groups
recent = df[df['release_year'] >= 2018]
old = df[df['release_year'] < 2018]

# Count TV shows
success_recent = (recent['type'] == 'TV Show').sum()
success_old = (old['type'] == 'TV Show').sum()

# Total samples
n_recent = len(recent)
n_old = len(old)

# Z-test
count = np.array([success_recent, success_old])
nobs = np.array([n_recent, n_old])

z_stat, p_value = proportions_ztest(count, nobs, alternative='larger')

print("Z-stat:", z_stat)
print("P-value:", p_value)

# Decision
alpha = 0.05
if p_value < alpha:
    print("Reject H0: TV Shows proportion increased in recent years")
else:
    print("Fail to Reject H0")
```

```
Z-stat: 17.28945184862103
P-value: 2.8243427889897262e-67
Reject H0: TV Shows proportion increased in recent years
```

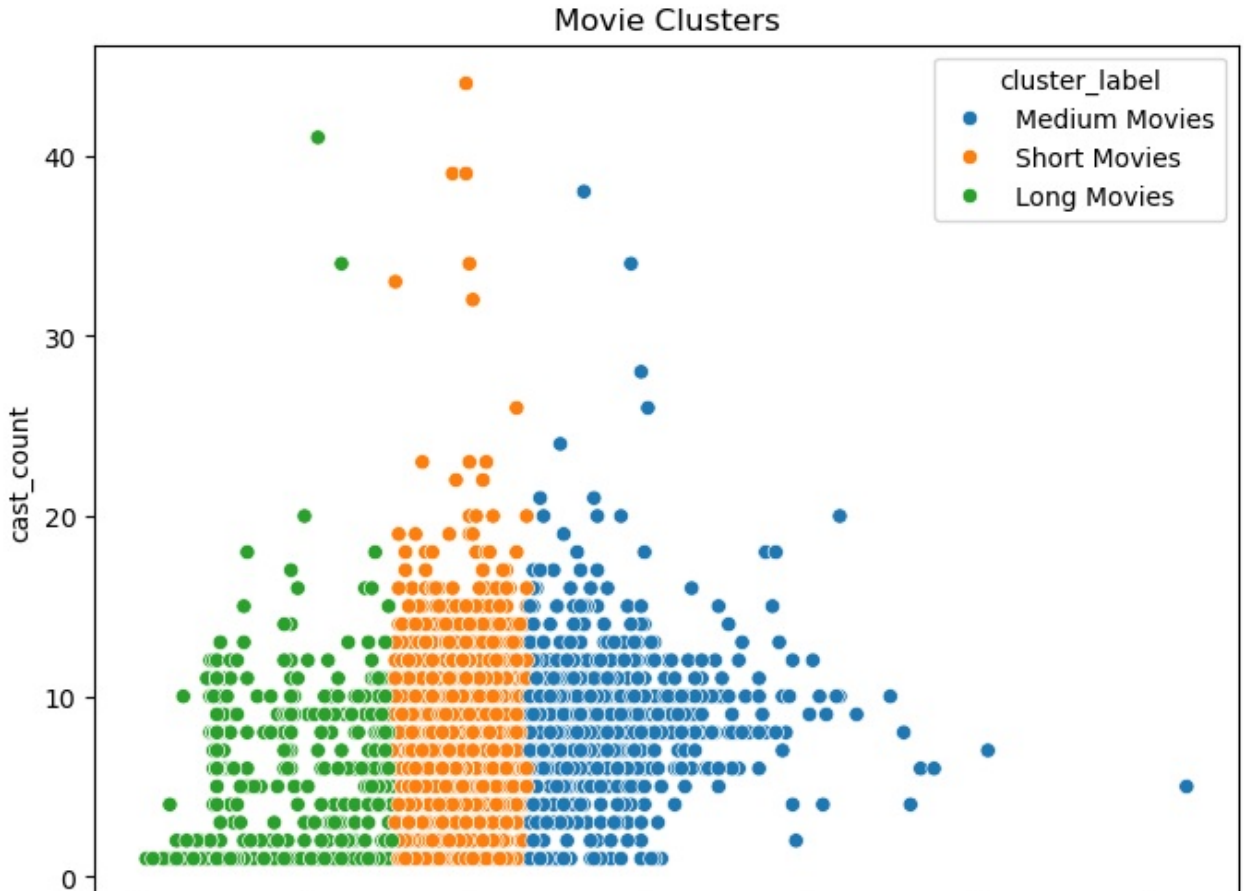
In [ ]:

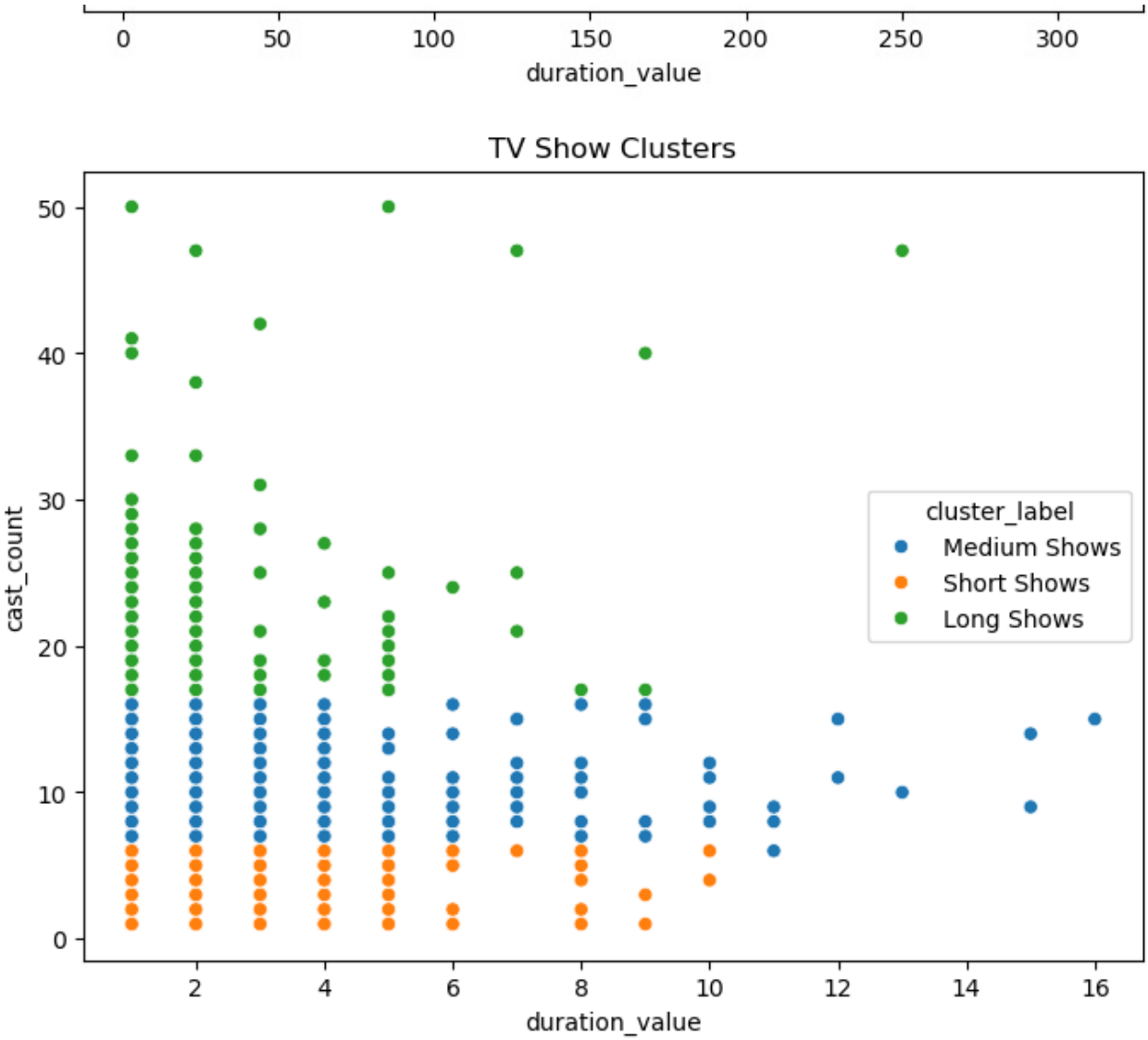
Q12] Can Netflix content be grouped into meaningful clusters based on features like genre, cast size, and duration?

In [56]:

```
from sklearn.cluster import KMeans



movies_df = df[df['type'] == 'Movie'].copy()
tv_df = df[df['type'] == 'TV Show'].copy()


features = ['duration_value', 'cast_count', 'genre_count']





kmeans_movies = KMeans(n_clusters=3, random_state=42)
movies_df['cluster'] = kmeans_movies.fit_predict(movies_df[features])

movie_labels = {
    0: "Short Movies",
    1: "Medium Movies",
    2: "Long Movies"
}
movies_df['cluster_label'] = movies_df['cluster'].map(movie_labels)
```

```
kmeans_tv = KMeans(n_clusters=3, random_state=42)
tv_df['cluster'] = kmeans_tv.fit_predict(tv_df[features])

tv_labels = {
    0: "Short Shows",
    1: "Medium Shows",
    2: "Long Shows"
}
tv_df['cluster_label'] = tv_df['cluster'].map(tv_labels)









# Movies Plot
plt.figure(figsize=(8,6))
sns.scatterplot(
    x='duration_value',
    y='cast_count',
    hue='cluster_label',
    data=movies_df
)
plt.title("Movie Clusters")
plt.show()

# TV Shows Plot
plt.figure(figsize=(8,6))
sns.scatterplot(
    x='duration_value',
    y='cast_count',
    hue='cluster_label',
    data=tv_df
)
plt.title("TV Show Clusters")
plt.show()
```





In [57]:

```
print("\n--- MOVIE CLUSTERS ---")
for label in movies_df['cluster_label'].unique():
    subset = movies_df[movies_df['cluster_label'] == label]
    print(label)
    print("Avg Duration:", subset['duration_value'].mean())
    print("Avg Cast:", subset['cast_count'].mean())

print("\n--- TV SHOW CLUSTERS ---")
for label in tv_df['cluster_label'].unique():
    subset = tv_df[tv_df['cluster_label'] == label]
    print(label)
    print("Avg Duration:", subset['duration_value'].mean())
    print("Avg Cast:", subset['cast_count'].mean())
```

```
--- MOVIE CLUSTERS ---
Medium Movies
Avg Duration: 136.6070549630845
Avg Cast: 8.71862182116489
Short Movies
Avg Duration: 96.86318332829961
Avg Cast: 7.569918453639384
Long Movies
Avg Duration: 55.184179456906726
Avg Cast: 3.5962219598583234

--- TV SHOW CLUSTERS ---
Medium Shows
Avg Duration: 1.9271012006861064
Avg Cast: 10.01114922813036
Short Shows
Avg Duration: 1.5527502254283139
Avg Cast: 2.8872858431018935
```

```
Long Shows
Avg Duration: 2.303703703703704
Avg Cast: 22.666666666666668
```

Q13] Which genres are overcrowded vs underserved?

In [58]:

```
from collections import Counter

new_df = df.copy()
new_df['genre_list'] = new_df['genres'].apply(lambda x: x.split(','))

all_genres = [g.strip() for sublist in new_df['genre_list'] for g in sublist]

genre_counts = Counter(all_genres)

genre_df = pd.DataFrame(genre_counts.items(), columns=['genre', 'count'])
genre_df = genre_df.sort_values(by='count', ascending=False)
```

In [59]:

```
print("Overcrowded genres:")
genre_df.head(10)
```

```
Overcrowded genres:
```

Out[59]:

	genre	count
4	International Movies	2437
6	Dramas	2106
8	Comedies	1471
0	International TV Shows	1199
15	Documentaries	786
13	Action & Adventure	721
1	TV Dramas	704
12	Independent Movies	673
21	Children & Family Movies	532
14	Romantic Movies	531

In [60]:

```
print("Underserved genres:")
genre_df.tail(10)
```

```
Underserved genres:
```

Out[60]:

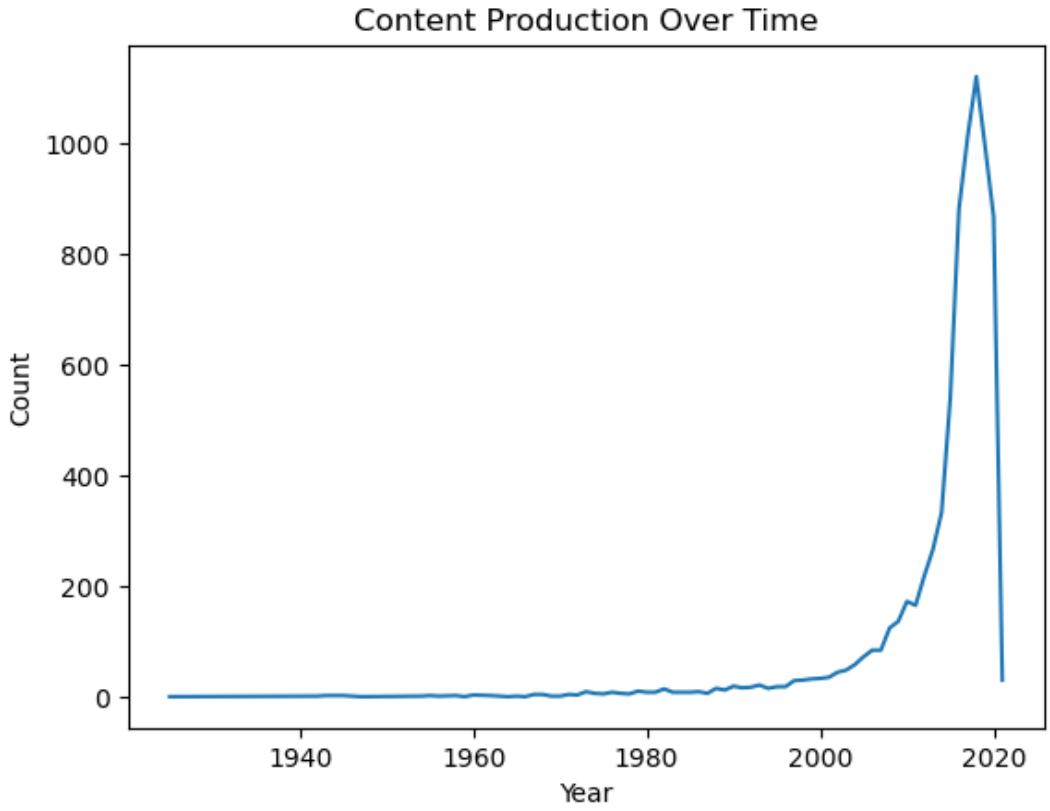
	genre	count
38	TV Horror	69
36	Teen TV Shows	60
37	Cult Movies	59
23	Faith & Spirituality	57
41	Anime Features	57
26	Movies	56
35	Stand-Up Comedy & Talk Shows	52

33	TV Thrillers	50
40	Classic & Cult TV	27
20	TV Shows	12

Q14] How has content production evolved over time?

In [61]:

```
import matplotlib.pyplot as plt

df['release_year'] = pd.to_numeric(df['release_year'], errors='coerce')

yearly = df.groupby('release_year').size()

yearly.plot(kind='line')
plt.title("Content Production Over Time")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```



CONCLUSION] With this we can say that most content-addition was done between the years 2000 and 2020.

In [ ]:

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In [ ]:

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